

Project 2: RAG Parameter Exploration

CourseCode: 515512

Group #: 17

Student ID: 112701018、113550020、112705068、111701030

Name: 姜明志、賴宇弘、何昀珍、程品諭

Please keep the title of each section and its questions; deleting any of them will lead to a penalty.

Task 1 Simple Baseline: LLM Only (6%)

1. (3%) Explain what kind of Prompt Engineering you try and describe your observation.

首先將預設的 prompt 依 RTCF 的格式整理成以下形式：

Please bring into the character described below:

Role: You are an expert on Arthur Conan Doyle's Sherlock Holmes stories.

Task: Answer the user's question directly, concisely, and accurately. Avoid adding any information that is not relevant to the question.

Context: Answer based purely on your internal knowledge of the Sherlock Holmes canon without external documents.

Format: A clear and brief textual response.

結果：發現多數的 Golden Answer 都比輸出還要短，所以更改了 Task 跟 Format 讓答案限制在至多 5 個單字

Please bring into the character described below:

Role: You are an expert on Arthur Conan Doyle's Sherlock Holmes stories.

Task: Answer the user's question with only the exact keyword or short phrase. Do not write full sentences nor provide explanations.

Context: Answer based purely on your internal knowledge of the Sherlock Holmes canon without external documents.

Format: Strictly maximum of 5 words.

結果:發現多數問題回答嚴重錯誤, 雖然在僅限 Prompt Engineering 的 Task 1 很難真正消除幻覺, 但這邊嘗試讓 LLM 在無法回答的情況下回答 “Unknown”

Please bring into the character described below:

Role: You are an expert on Arthur Conan Doyle's Sherlock Holmes stories.

Task: Answer the user's question with only the exact keyword or short phrase. Do not write full sentences nor provide explanations. If you cannot give an exact answer, simply reply "Unknown".

Context: Answer based purely on your internal knowledge of the Sherlock Holmes canon without external documents.

Format: Strictly maximum of 5 words, or "Unknown" if you cannot give an exact answer.

結果:回答中沒有“Unknown”的出現, 但是部分答案變成是正確的

最後的 Answer / Golden Answer 如下:

QID	Answer	Golden Answer
1	Chess	About education
2	Indian Army	India
3	Clients	Toby
4	Sherlock Holmes	Sherlock Holmes
5	Raven	Ritual
6	Dreams	Knowledge of Anatomy
7	Young's ultimatum	Marry a Mormon elder's son within one month or be penalized.
8	Financial irregularities, gambling debts.	Poverty/pawn habits and heavy drinking.
9	Misdirection, deception, and profit.	To prevent others from enjoying wealth.
10	Small-bore rifle.	A heavy cleaver

2. (3%) **Provide examples** to demonstrate what kind of queries get performance improvement through prompt engineering.

例子1: 答案格式簡單的問題

這類問題可以透過修改 Format 限制回答字數來解決, 由於答案簡短所以能回答出的答案有將變得更有限, 也就更有可能有更好的結果。

以 Q2 為例:

Golden Answer: India

Answer: Mary's father was a captain in the Northumberland Regiment.

→ Indian Army

例子2: 發生過度解答的情況

可以修改 Task 限制不要過度解釋或重複問題中的文字, 僅回答相應的答案即可。但答案有可能從 "與正解類似但過度解答" 變成 "偏離正解但精簡回答", 所以這邊的優化僅限於滿足回答的形式。

以 Q10 為例:

Golden Answer: A heavy cleaver

Answer: A small, sharp-pointed instrument, likely a penknife, was used to sever Hatherley's thumb.

→ Small-bore rifle.

Task 2 Medium: BM25 + LLM (14%)

1. (2%) Briefly describe what happened after adding retrieval into the system.

結果對比：

	Task 1 (LLM Only)	Task 2 (BM25 + LLM)
答對題數	1/10 (QID 4)	4/10 (QID 1, 5, 7, 9)
主要失敗原因	幻覺	關鍵詞情境錯誤、關鍵詞缺失
答案風格	簡短精煉	完整的句子

在加入 BM25 retrieval 之後，總答對題數從 1/10 提升至 4/10。

在 Task 1 中雖然使用了 prompt engineering 限制輸出格式，但在大多數題目中 LLM 仍產生了嚴重的幻覺情形(例如 QID 1 回答 "Chess"、QID 3 回答 "Clients"、QID 5 回答 "Raven" 等等)，顯示出了在缺乏外部資料的情況下，模型容易以訓練的記憶捏造看似合理的答案。

而 Task 2 透過 BM25 提供相關段落作為參考，有效抑制了部分的幻覺情況。尤其是在包含稀有的專有名詞問題上改善較為顯著，因為 BM25 的 IDF 機制會給予稀有詞更高的檢索權重，最後提升了正確 chunk 的排名。

但我觀察到 Task 2 仍無法解決的失敗類型主要有以下兩種：

- 情境錯誤：相同關鍵詞在多個故事出現，BM25 無法辨別正確情境(例如 QID 2, 4, 8, 10)。
- 關鍵詞缺失：答案未出現在問題關鍵詞中，BM25 無法匹配(例如 QID 3 的 "Toby")。

2. (6%) Hold **k₁** value fixed while adjusting **b** value, show how **two queries'** answers changed from incorrect to correct. Please include **screenshots(query & answer)** of your answers, and discuss why these differences occur.

- **BM25Okapi(tokenized_chunks, k1=1.5, b=0.0)**

```
QID: 1
Prompt:
System: You are an expert on Arthur Conan Doyle's Sherlock Holmes stories.
Based on the user's question, identify the relevant information from the provided Sherlock Holmes passage
excerpts.
Answer directly, concisely, and accurately.
Do not include information that is unrelated to the question.
If the provided excerpts do not contain enough evidence to answer the question, say that the answer cannot be
determined from the given passages.

User: ### Question
About which of these things does Sherlock Holmes say, "It is a series of lessons, with the greatest for the
last."

### Sherlock Holmes passage excerpts
{ relevant_chunks }

Answer:
According to the passage, Sherlock Holmes says, "It is a series of lessons, with the greatest for the last."

Golden Answer:
About education

Relevant Chunks:
['"Education never ends, Watson. It is a series of lessons with the greatest for the last. This is
an instructive case. There is neither money nor credit in it, and yet one would wish to tidy it up.
When dusk comes we should find ourselves one stage advanced in our investigation." When we returned to Mrs.
Warren's rooms, the gloom of a London winter evening had thickened into one gray curtain, a dead monotone of
colour, unbroken only by the sharp yellow squares of the windows and the blurred haloes of the gas-lamps']
```

```
QID: 3
Prompt:
System: You are an expert on Arthur Conan Doyle's Sherlock Holmes stories.
Based on the user's question, identify the relevant information from the provided Sherlock Holmes passage
excerpts.
Answer directly, concisely, and accurately.
Do not include information that is unrelated to the question.
If the provided excerpts do not contain enough evidence to answer the question, say that the answer cannot be
determined from the given passages.

User: ### Question
What was the name of the dog Sherlock Holmes uses to find the men he is seeking in 'The sign of the four?'

### Sherlock Holmes passage excerpts
{ relevant_chunks }

Answer:
The name of the dog is not explicitly stated in the provided excerpts. The passage mentions "the curious
incident of the dog in the nighttime," but it does not reveal the dog's name.

Golden Answer:
Toby

Relevant Chunks:
['ear boy! it was simplicity itself. I don't wish to be theatrical. It is all patent and above-board. Two
officers who are in command of a convict-guard learn an important secret as to buried treasure. A map is drawn
for them by an Englishman named Jonathan Small. You remember that we saw the name upon the chart
in Captain Morstan's possession. He had signed it in behalf of himself and his associates, the sign of the
four, as he somewhat dramatically called it. Aided by this chart, the officers—or one of them']
```

- **BM25Okapi(tokenized_chunks, k1=1.5, b=1.0)**

```
QID: 1
Prompt:
System: You are an expert on Arthur Conan Doyle's Sherlock Holmes stories.
Based on the user's question, identify the relevant information from the provided Sherlock Holmes passage
excerpts.
Answer directly, concisely, and accurately.
Do not include information that is unrelated to the question.
If the provided excerpts do not contain enough evidence to answer the question, say that the answer cannot be
determined from the given passages.

User: ### Question
About which of these things does Sherlock Holmes say, "It is a series of lessons, with the greatest for the
last."

### Sherlock Holmes passage excerpts
{ relevant_chunks }

Answer:
Sherlock Holmes says, "It is a series of lessons, with the greatest for the last." in the first excerpt.

Specifically, he states this in the first paragraph: "Education never ends, Watson. It is a series of lessons
with the greatest for the last."

Golden Answer:
About education

Relevant Chunks:
['\n"Education never ends, Watson. It is a series\nof lessons with the greatest for the last. This is
an\ninstructive case. There is neither money nor credit\nin it, and yet one would wish to tidy it up.
When\ndusk comes we should find ourselves one stage\nadvanced in our investigation."\n\nWhen we returned to Mrs.
Warren's rooms, the\ngloom of a London winter evening had thickened\ninto one gray curtain, a dead monotone of
colour,\nbroken only by the sharp yellow squares of the\nwindows and the blurred haloes of the gas-l']
```

```
QID: 3
Prompt:
System: You are an expert on Arthur Conan Doyle's Sherlock Holmes stories.
Based on the user's question, identify the relevant information from the provided Sherlock Holmes passage
excerpts.
Answer directly, concisely, and accurately.
Do not include information that is unrelated to the question.
If the provided excerpts do not contain enough evidence to answer the question, say that the answer cannot be
determined from the given passages.

User: ### Question
What was the name of the dog Sherlock Holmes uses to find the men he is seeking in 'The sign of the four?

### Sherlock Holmes passage excerpts
{ relevant_chunks }

Answer:
The passage excerpt does not mention the name of the dog Sherlock Holmes uses to find the men he is seeking in
'The Sign of the Four.' It does, however, describe a "curious incident of the dog in the nighttime."

Golden Answer:
Toby

Relevant Chunks:
['ear boy! it was simplicity itself.\nI don't wish to be theatrical. It is all patent and\nabove-board. Two
officers who are in command\nof a convict-guard learn an important secret as\nto buried treasure. A map is drawn
for them by\nan Englishman named Jonathan Small. You re-\nmember that we saw the name upon the chart
in\nCaptain Morstan's possession. He had signed it\nin behalf of himself and his associates,-the sign\nof the
four, as he somewhat dramatically called\nit. Aided by this chart, the officers-or one of\nthem']
```

實驗 $k1=1.5$ (固定預設值) , 分別測試 $b=0.0$ 和 $b=1.0$ 。

b 主要控制文件長度的正規化程度, $b=0$ 表示完全不考慮長度差異; $b=1$ 為完全按比例懲罰長文件, 預設的 $b=0.75$ 為適度懲罰。

最終結果:

QID	$b=0.0$	$b=1.0$	變化
1	答對	答對	無變化
3	答錯	答錯	相同 chunk, 相同錯誤

個人觀察:

改變 b 對所有題目的檢索結果幾乎沒有太多影響, 總答對題數皆為 3/10 (和一開始預設的 $k1=1.5$, $b=0.75$ 相同)。猜測主要的原因是資料集使用固定 $chunk_size=512$ 和 $stride=512$ 進行切割, 因此所有 chunk 長度幾乎一致, 長度正規化的效果接近常數, 導致不影響 chunk 之間的相對排名。

因此從觀察得出, b 參數的效果在 uniform-length chunking 下被消除。若使用 variable-length chunking (例如依照段落進行切割), b 的影響可能才會顯現。

因為從結果上沒有看出 b 參數改變的影響, 最後選用 $b=0.75$ 的預設值。

3. (6%) Hold **b value fixed** while adjusting **k₁ value**, show how **two queries'** answers changed from incorrect to correct. Please include **screenshots(query & answer)** of your answers, and discuss why these differences occur.

- **BM25Okapi(tokenized_chunks, k1=0.5, b=0.75)**

```
QID: 1
Prompt:
System: You are an expert on Arthur Conan Doyle's Sherlock Holmes stories.
Based on the user's question, identify the relevant information from the provided Sherlock Holmes passage
excerpts.
Answer directly, concisely, and accurately.
Do not include information that is unrelated to the question.
If the provided excerpts do not contain enough evidence to answer the question, say that the answer cannot be
determined from the given passages.

User: ### Question
About which of these things does Sherlock Holmes say, "It is a series of lessons, with the greatest for the
last."

### Sherlock Holmes passage excerpts
{ relevant_chunks }

Answer:
Sherlock Holmes says, "It is a series of lessons, with the greatest for the last." in the first excerpt, the
one that begins with "Education never ends..."

Golden Answer:
About education

Relevant Chunks:
['\n"Education never ends, Watson. It is a series\nof lessons with the greatest for the last. This is
an\ninstructive case. There is neither money nor credit\nin it, and yet one would wish to tidy it up.
When\ndusk comes we should find ourselves one stage\nadvanced in our investigation."\nWhen we returned to Mrs.
Warren's rooms, the\ngloom of a London winter evening had thickened\ninto one gray curtain, a dead monotone of
colour,\nbroken only by the sharp yellow squares of the\nwindows and the blurred haloes of the gas-l']
```

```
QID: 3
Prompt:
System: You are an expert on Arthur Conan Doyle's Sherlock Holmes stories.
Based on the user's question, identify the relevant information from the provided Sherlock Holmes passage
excerpts.
Answer directly, concisely, and accurately.
Do not include information that is unrelated to the question.
If the provided excerpts do not contain enough evidence to answer the question, say that the answer cannot be
determined from the given passages.

User: ### Question
What was the name of the dog Sherlock Holmes uses to find the men he is seeking in 'The sign of the four?

### Sherlock Holmes passage excerpts
{ relevant_chunks }

Answer:
The passage does not mention the name of Sherlock Holmes's dog. It describes a sheep-dog of the moor and a
"spectral hound."

Golden Answer:
Toby

Relevant Chunks:
['e were seeking. Holmes knocked\nwithout receiving any answer, and then tried to\nturn the handle and force it
open. It was locked\non the inside, however, and by a broad and pow-\nerful bolt, as we could see when we set
our lamp\nup against it. The key being turned, however, the\nhole was not entirely closed. Sherlock Holmes
bent\ndown to it, and instantly rose again with a sharp\nintaking of the breath.\n"There is something devilish
in this, Watson,"\nsaid he, more moved than I had ever before seen\nhim. "What do you']
```


- **BM25Okapi(tokenized_chunks, k1=3.0, b=0.75)**

```
QID: 1
Prompt:
System: You are an expert on Arthur Conan Doyle's Sherlock Holmes stories.
Based on the user's question, identify the relevant information from the provided Sherlock Holmes passage
excerpts.
Answer directly, concisely, and accurately.
Do not include information that is unrelated to the question.
If the provided excerpts do not contain enough evidence to answer the question, say that the answer cannot be
determined from the given passages.

User: ### Question
About which of these things does Sherlock Holmes say, "It is a series of lessons, with the greatest for the
last."

### Sherlock Holmes passage excerpts
{ relevant_chunks }

Answer:
Sherlock Holmes says, "It is a series of lessons, with the greatest for the last." in the first paragraph of
the passage.

Golden Answer:
About education

Relevant Chunks:
['\n"Education never ends, Watson. It is a series\nof lessons with the greatest for the last. This is
an\ninstructive case. There is neither money nor credit\nin it, and yet one would wish to tidy it up.
When\ndusk comes we should find ourselves one stage\nadvanced in our investigation."\nWhen we returned to Mrs.
Warren's rooms, the\ngloom of a London winter evening had thickened\ninto one gray curtain, a dead monotone of
colour,\nbroken only by the sharp yellow squares of the\nwindows and the blurred haloes of the gas-l']
```

```
QID: 3
Prompt:
System: You are an expert on Arthur Conan Doyle's Sherlock Holmes stories.
Based on the user's question, identify the relevant information from the provided Sherlock Holmes passage
excerpts.
Answer directly, concisely, and accurately.
Do not include information that is unrelated to the question.
If the provided excerpts do not contain enough evidence to answer the question, say that the answer cannot be
determined from the given passages.

User: ### Question
What was the name of the dog Sherlock Holmes uses to find the men he is seeking in 'The sign of the four?'

### Sherlock Holmes passage excerpts
{ relevant_chunks }

Answer:
The passage does not mention the name of the dog Sherlock Holmes uses. It discusses Sherlock Holmes's
unconventional methods of deduction, particularly his use of dogs in investigations, and mentions a case
involving a map called "the sign of the four."

Golden Answer:
Toby

Relevant Chunks:
['\n"To the curious incident of the dog in the night-\ntime."\n"The dog did nothing in the night-time."\n"That
was the curious incident," remarked\nSherlock Holmes.\nFour days later Holmes and I were again in the\ntrain,
bound for Winchester to see the race for the\nWessex Cup. Colonel Ross met us by appointment\noutside the
station, and we drove in his drag to the\ncourse beyond the town. His face was grave, and\nhis manner was cold
in the extreme.\n"I have seen nothing of my horse," said he.\n"I suppose that you would k']
```

實驗 $b=0.75$ (固定預設值) , 分別測試 $k_1=0.5$ 和 $k_1=3.0$ 。

k_1 控制詞頻(TF)的飽和速度。當 k_1 越小, 出現一次和出現多次的詞分數差距越小(因為較快速飽和), 而 k_1 越大, 高頻詞的分數越高。

最終結果:

QID	$k_1=0.5$	$k_1=3.0$	變化
1	答對	答對	無變化
3	答錯	答錯	不同 chunk, 但仍答錯

個人觀察:

調整 k_1 對多數題目的檢索結果幾乎沒有影響, 總答對題數皆為 4/10 (皆比一開始預設的 $k_1=1.5$, $b=0.75$ 多答對一題)。猜測可能的原因是資料集所有 chunk 的長度固定 (512 tokens), 因此各 chunk 詞頻分布差距不大。BM25 排名主要由 IDF 決定, 而 IDF 不受 k_1 影響。

值得注意的是 QID 3: 兩個 k_1 值都無法找到含有 "Toby" 的正確 chunk, 這是因為問題本身根本沒有出現 "Toby" 這個關鍵詞, 所以 BM25 無法進行詞彙匹配, 為 BM25 的根本限制 (關鍵詞缺失)。

最後選用 $k_1=0.5$, 使系統更依賴 IDF, 以符合固定長度 chunk 的資料特性。

Task 3 Strong: embedding + LLM (20%)

1. (5%) Show how you modify your prompt design after adding retrieval, and **explain** your strategy.

- Modified Prompt Design

```
SYSTEM_PROMPT = """
```

```
Please bring into the character described below:
```

```
Role: You are a world-class expert on Arthur Conan Doyle's  
Sherlock Holmes canon and its well-known adaptations.
```

```
Task: Answer the question using the retrieved excerpts together  
with your own knowledge. Follow these steps IN ORDER:
```

```
    1. Decide EXACTLY what the question asks for (a name, a title,  
    a place, a date, or a short phrase).
```

```
    2. Use an excerpt ONLY if it clearly concerns the SAME story /  
    person / thing the question asks about; IGNORE excerpts about  
    anything else.
```

```
    3. If no excerpt fits, answer from your OWN knowledge of the  
    canon. NEVER let a noisy excerpt override a fact you know.
```

```
    4. Commit to ONE answer. Do NOT list alternatives and do NOT  
    write explanations.
```

```
Context: The retrieved excerpts are NOISY – they are often from a  
DIFFERENT story, and several are irrelevant.
```

```
Format: Put the answer on the FIRST line, as SHORT as possible.  
If you are genuinely unsure, reply with EXACTLY: I don't know
```

```
""".strip()
```

```
USER_PROMPT = """
```

```
### Question
```

```
{query}
```

```
### Retrieved Sherlock Holmes passages (may be noisy or from a  
different story)
```

```
{relevant_chunks}
```

```
""".strip()
```

- Explanation

在 Task 1 (LLM only) 階段，我們採用 RTCF (Role–Task–Context–Format) 框架設計 Prompt，要求模型僅依靠自身知識作答，並且 Context 明確限制模型必須回答 Without external documents，而 Format 則要求輸出答案長度不超過五個字。

而在 Task 3 (Embedding + LLM) 階段，系統會先透過 Embedding retrieval 檢索出相關段落，再交由 LLM 作答，所以我們在設計 Prompt 時，除了要關注回答問題的方式，更需要有效運用 Retrieved excerpts，同時避免受到 Retrieval noise 干擾。

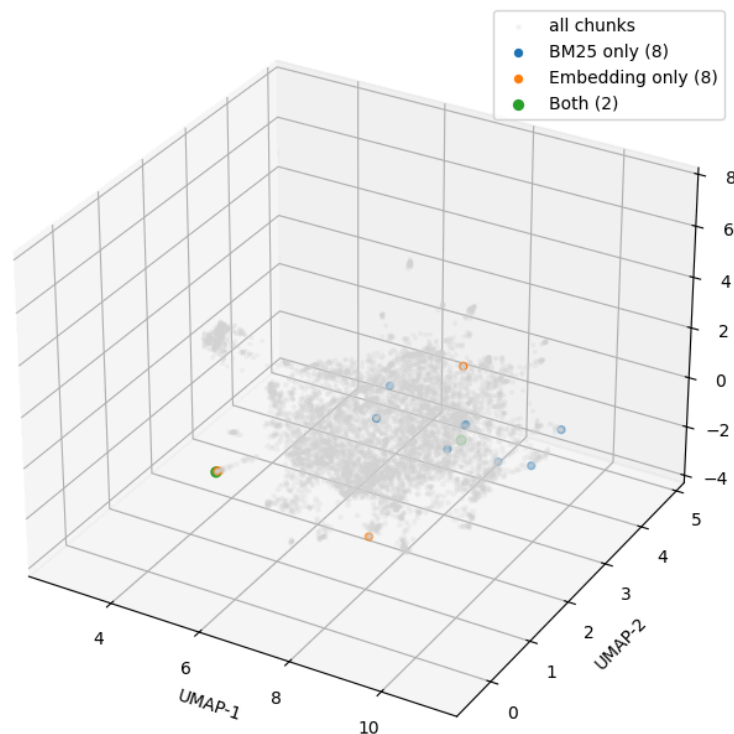
因此，我們以 Task 1 的 RTCF 架構為基礎，針對各個欄位進行優化和改進：

1. 針對 Role 欄位，將原先「Sherlock Holmes expert」擴充為「Canonical works and major adaptations expert」，藉此涵蓋 Questions Dataset (final_query.jsonl) 中，有部分題目並非出自原著，而是源自改編的影視作品。
2. 針對 Task 欄位，改進為採取 IN ORDER 為核心的 Decision process:「先判斷題目需求，其次檢查 Retrieved excerpt 是否和 Question 來自同篇 Story 或同個 Topic，倘若 Excerpt 與 Question 不相符，則優先根據自身知識作答，並在最後，只輸出一個明確答案」。透過將作答流程拆解為多個步驟，期望模型能夠主動辨識 Retrieval noise，而非直接將所有檢索到的內容作為可靠證據。
3. 而 Task 3 階段最核心的修正部分，便是針對 Context 欄位，相較於 Task 1 的 Without external documents 設定，我們明確告知 Model 將收到 Retrieved excerpts，但這些 Excerpts 可能包含 Noise、其他低關聯的 Stories，甚至與題目完全無關的內容。因此，限制 Model 只在 Excerpt 與 Question 具有明確主題關聯時，才能採用 Excerpt，否則忽略檢索結果，並改為以既有知識作答。藉此，避免 Model 受到來自其他 Stories 的人物、地點或事件干擾，導致答案偏離正確篇章。
4. 再來，針對 Format 部分，則根據 Evaluation rule 進行調整，將 Abstention string 從 Unknown 改為 RAG Judge System 能正確辨識的字串「I don't know」，並且移除「答案不得超過五字」的限制，因為部分 Ground-truth Answer 便超過 5 個字，避免正確答案被截斷。

與此同時，在設計 Prompt 時，也適度使用大寫關鍵字，比如：「EXACTLY、ONE、IGNORE、NEVER、OWN、NOISE」，藉此強調重要的 Instruction，包括：輸出單一答案、避免受到 Noise 影響，以及確保 Model 使用正確的 Abstention string。

2. (10%) Using chunk size=512, compare the most relevant 10 chunks between BM25 and embedding in the semantic space, apply UMAP for dimensionality reduction to compress the chunk vectors into 3D. **Provide a screenshot** of plotting the retrieved chunks in a 3D space and color-coded them to visualize the differences between BM25 and embedding. **(You must use colab to plot it) And describe your observation. You can use any query to finish this part.**

Retrieved Top 10 Chunks for Query ID 4 (BM25 vs. Embedding)



- 先取 Query #4 的 BM25 top-10 與 Embedding top-10, 再透過 UMAP 將全部 Chunk 的 Embedding 向量降到 3 維, 並在 Semantic Space 中以不同顏色標示, 包括: 灰色代表全部 Chunks、藍色圓點代表僅 BM25 取到、橘色圓點代表僅 Embedding 取到、綠色圓點代表兩者都有取到。
- 從圖表中可以發現, 兩種檢索方式, 幾乎沒有重疊。綠色圓點 (both) 只有 2 個, BM25 與 Embedding 各自有 8 個獨有 Chunk, 代表兩種方法各自取回 10 個 Chunk, 其中僅有 2 個重疊 (20%)。
- 另外, 由於 UMAP 在降維時, 會盡可能保留資料原本語義鄰近的關係, 所以圖中點跟點距離越近, 說明兩者的語義越相近。相對的, 兩者位置越分散, 則代表語義來源越多元。而圖中大量灰色點集中的區域, 則反映文章內容主要集中的 Semantic Space。
- 與此同時, BM25 取到的 Chunk (藍色圓點) 主要集中在 Chunk 密度高的區域。而 Embedding 取到的 Chunk (橘色圓點) 則明顯更分散, 除了有落在 Chunk 密度高的區域, 也有數個點落在更加離群的外側位置, 整體分布範圍比 BM25 更廣。
- 而造成 Embedding 取到的 Chunk (橘色圓點) 明顯更分散的差距現象原因, 在於 Embedding 本身便是進行 Semantic 比對, 它檢索到的內容, 包含意思相關但用字不同的 Chunk, 來源相較 BM25 會更加廣泛, 所以分散在不同區域。

- 除此之外, BM25 和 Embedding 兩者檢索方法的重疊範圍極小 (2/10), 說明彼此檢索內容高度互補。BM25 能夠檢索到字面關鍵詞, 集中在語義密集的主要段落; 而 Embedding 能夠補上語義相關, 但用字不同、分佈更廣的段落。這也說明 Task 4 以 RRF 融合兩種檢索方法的必要性:「單一種檢索方法都有各自盲點, 倘若只仰賴一種檢索方式, 都會漏掉其他有價值的 Chunk, 而藉由 RRF 融合, 能同時兼顧字面和語義兩個維度的內容」。

3. (5%) Please experience different chunk sizes, chunk overlaps to observe the influence of the results. Fill in the table below to show your experiments' results. Feel free to add more rows.

Chunk size	Chunk overlap	Give the retrieval result a score	Rationale
256	0 (0%)	4/10	透過對 sample_query.jsonl 的 10 題 Query, 使用 Embedding Model 取 Top-10 Chunk, 檢查 Gold Answer 關鍵詞是否全部出現在 Chunk 原文, 判斷在送入 LLM 之前, 有沒有把回答問題需要的原文帶進來。另外, 由於僅評估 Embedding Retrieval 覆蓋率, 實際 Pipeline 採用 BM25 + Embedding RRF 融合, 整體覆蓋率通常更高。 而經過實驗發現, 由於 Chunk 僅有 256 字, 並且完全沒有 Overlap, 文章會被切成許多獨立片段, Golden Answers 經常橫跨兩個 Chunk 邊界, 導致前半段落落在這一塊, 後半段則落在下一塊, 導致任何單一 Chunk 都無法完整涵蓋答案資訊。 因此, Retrieval 階段找到完整證據的機率會降低, 導致 10 題中僅 4 題成功命中。
256	128 (50%)	4/10	同樣採用 256 字的小型 Chunk, 檢索階段 10 題中仍有 4 題成功命中, 然而, 50% Overlap 讓相鄰 Chunk 彼此能共享一半內容, 相當於以滑動視窗(Sliding window)方式重新切分文本, 能降低答案被邊界切斷的機率。 雖然從檢索指標沒有明顯提升, 但由於每個 Chunk 內容集中、雜訊較少, 能更容易聚焦在真正相關的資訊, 並且較短的 Chunk 也代表放入 Prompt 的 Tokens 數量較少, 送入 LLM 後, 能更容易聚焦在真正相關的資訊。
256	192 (75%)	4/10	將 Overlap 從 50% 提高到 75% 後, 理論上能進一步降低答案被切斷的風險, 但 10 題中依然僅 4 題成功命中。然而, 過高的 Overlap 也會讓 Chunk 數量從約 28000 個增加至約 56000 個, 導致建立 Embedding 和建構索引的成本增加。

384	288 (75%)	5/10	相較於 256 字 Chunk, 將字數提升至 384 字, 能夠在單一片段中, 保留更多上下文資訊, 並且維持高 Overlap 的切分方式, 答案被邊界切斷的問題會更少, 能兼顧檢索能力和回答品質, 而10題中有 5 題成功命中。
512	0 (0%)	4/10	作為預設的 Chunking 策略 Baseline, Chunk 大小中等, 為 512 字, 能夠覆蓋一定的上下文內容, 並且沒有 Overlap 設定。然而, 在檢索能力或回答品質方面, 並沒有特別突出的成績, 對應的 10 題中有 4 題命中, 組合缺乏明顯優勢。
1024	0 (0%)	7/10	當 Chunk 增加到 1024 字後, 每個片段能涵蓋更多內容, 使得答案被包含於檢索結果的機率大幅提升, 因此, 在 10 題 Sample Query 中, 有 7 題成功命中, 為所有 Chunking 組合中命中數最高的設定。 然而, 這也可能提升單一 Chunk 夾帶與 Query 無關資訊的風險, 增加 Retrieval noise, 導致 LLM 在閱讀內容時, 可能更容易受到干擾, 即便有檢索到答案的上下文段落, 也未必能聚焦於正確資訊, 實際 Test 正確率未必會這麼高。
2048	1024 (50%)	7/10	2048 字搭配 Overlap, 能進一步提高文本覆蓋範圍, 所以在 10 題 Sample Query 中, 同樣維持成功命中 7 題的水準。 然而, 超大的 Chunk 帶來的疑慮, 與 1024 字設定相同, 雖然在 Retrieval 階段能將正確資訊納入, 但也會引入大量無關的內容, 可能讓後續 LLM 在實際推理時, 受到更嚴重的 Noise 影響。 因此, 綜合以上所有實驗組合, Chunk 越大, Embedding Retrieval 覆蓋率越高, 但送入 LLM 的 Context Noise 也可能隨之增加; Overlap 越高, 答案被邊界切斷的風險越低。從中, 可以了解並非單一指標越高越好, 還需要兼顧覆蓋率、雜訊控制, 找到最適合的平衡點。

Task 4 Hybrid: LLM+BM25+embedding (30%)

1. (5%) Describe how you integrate the results from retrieval methods.

用RRF做late fusion:

1. 分別從bm25, embedding得到兩條topk的獨立排名
2. 以排名融合, 對每個 chunk d,

$$\text{score}(d) = \sum_{\text{retriever}} \frac{1}{k + \text{rank}_{\text{retriever}}(d)}, \quad k = 60$$

融合分數排序取前 20作為POOL。

3. 在 RRF 融合前後, 系統會做三層優先級較高的檢查, 用來覆蓋或限縮演算法的結果, 確保精確度 (Precision):

-**Named-story Restrict**: 題目點名某故事時, 把候選限制在該故事 chunks (外部 KB 仍以 restrict | set(EXT_IDXS) 保持可被檢索);

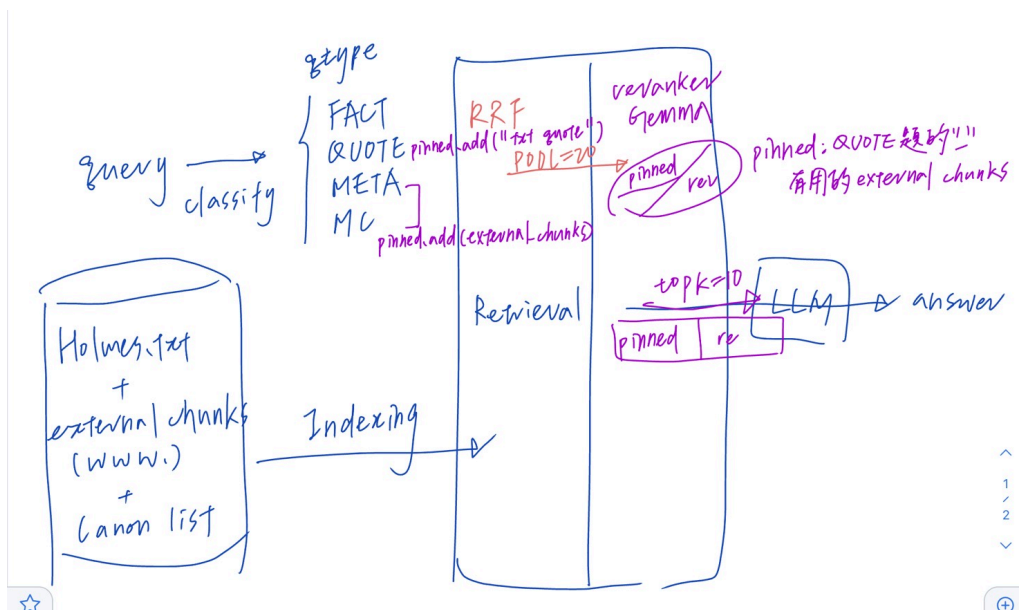
-**Quote Anchoring**: 題目引號定位回正典的段落被置頂;

-**External KB Flooring / Pin**: 若題型是META、FACT, 將前幾個相關的百科 chunk 補進POOL

4. 最後交給reranker(gemma)從 20 個候選選出 top-k=10 (pinned 槽位保留), 才送進LLM生成。

2. (5%) **Provide at least ONE example** query from **sample_query.jsonl** to demonstrate how your hybrid retrieval improves performance.

Design of our retrieval:



QID: 3

Prompt:

System: Please bring into the character described below:

Role: You are a world-class expert on the Sherlock Holmes canon.

Task: Answer using the retrieved excerpts. Each excerpt is prefixed with [Story: ...]; trust only excerpts that clearly concern the SAME story / person / thing the question asks about; ignore the rest.

Format: output ONLY the canonical answer on ONE line (a name, a title, a place, a number, or one word). Never copy a whole sentence, and NEVER answer with a word that is merely echoed from the question (e.g. the quoted title word, or a place / character already named in the question). Rules:

1. NAME questions ("what is the name of...", arch-enemy/nemesis, inspector, client, address): output the proper noun (e.g. "Moriarty", "Lestrade", "221B Baker Street"). If an excerpt only describes the person/thing, give the actual name from your own canon knowledge, not the description.
2. ABOUT / TO WHOM questions: the answer is the person named or addressed in the quote – usually a name right after a comma at the end (e.g. "..., Watson" -> Watson) – NOT the topic word of the sentence.
3. FILL-IN-THE-BLANK (a title or quote containing ____): output ONLY the missing word(s); never repeat a word that already appears in the question (e.g. "The ____ of the Baskervilles" -> "Hound").
4. DEFINITION ("what was X?"): give the definition sitting NEXT TO X, not X itself ("baritsu" -> "a Japanese system of wrestling"; "the speckled band" -> "a snake (a swamp adder)").
5. IN WHICH story/novel/work something happens or appears: output a canon story TITLE only (e.g. "The Final Problem") – never a place, character, or event that is named in the question, and never a narrative sentence.
6. HOW OLD / HOW MANY / WHAT YEAR / HOW LONG: output only the number, age, or year (e.g. "seven years older", "1881").
7. If the excerpts do not contain it, answer from your own canon knowledge. If truly unsure, reply exactly: I don't know

User: ### Question

What was the name of the dog Sherlock Holmes uses to find the men he is seeking in 'The sign of the four?

Retrieved Sherlock Holmes passages(may be noisy or from a different story)

[Story: External] The Hound of the Baskervilles is one of the four Sherlock Holmes novels; its full title completes as 'The Hound of the Baskervilles'.

[Story: The Sign of the Four] What was it that Mr. Sherlock Holmes wanted, sir?"

"He wanted a dog of yours."

"Ah! that would be Toby."

"Yes, Toby was the name."

"Toby lives at No. 7 on the left here." He moved slowly forward with his candle among the queer animal family which he had ga

[Story: The Sign of the Four] little short wi' you at first, for I'm guyed at by the children, and there's many a one just comes down this lane to knock me up. What was it that Mr. Sherlock Holmes wanted, sir?"

"He wanted a dog of yours."

"Ah! that would be Toby."

"Yes, Toby was the name

[Story: External] The four Sherlock Holmes novels are: A Study in Scarlet (1887), The Sign of the Four (1890), The Hound of the Baskervilles (1902), and The Valley of Fear (1915).

[Story: The Sign of the Four] e matter with the dog?"

growled Holmes. "They surely would not take a cab, or go off in a balloon."

"Perhaps they stood here for some time," I suggested.

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The Sign of the Four

"Ah! it's all right. He's off again," said my companion, in a tone of relief

[Story: The Sign of the Four] revices left were worn down and rounded upon the lower side, as though they had frequently been used as a ladder. Holmes clambered up, and, taking the dog from me, he dropped it over upon the other side.

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The Sign of the Four

"There's the print of wooden

[Story: External] In 'The Sign of the Four', Mary Morstan received one large lustrous pearl every year for six years before she came to consult Sherlock Holmes.

[Story: The Sign of the Four] is where he put his foot in getting out.

If you can trace him, I should think that Toby will have no difficulty. Now run down-stairs, loose the dog, and look out for Blondin."

By the time that I got out into the grounds Sherlock Holmes was on the roof, and

[Story: The Sign of the Four] retired life, and have no friends whom

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The Sign of the Four

I could appeal to. If I am here at six it will do, I suppose?"

"You must not be later," said Holmes. "There is one other point, however. Is this handwriting the same as that upon the pearl-box

[Story: The Sign of the Four] me. I have told you that I had picked up something of medicine. One day when Dr. Somerton was down with a fever a little Andaman Islander was picked

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The Sign of the Four

up by a convict-gang in the woods. He was sick to death, and had gone to a lonely

Answer:

Toby

Golden Answer:

Toby

Relevant Chunks:

["[Story: External] The Hound of the Baskervilles is one of the four Sherlock Holmes novels; its full title completes as 'The Hound of the Baskervilles'."]

我們比較三個retrieval 方式, 分別是vanilla BM25, vanilla Embedding, RRF

method	正解段落排在第幾名
BM25	#8
Embedding	#5
RRF	#2

Query analyzing: 本題qtype為FACT, 用FACT_system prompt檢索

本題的關鍵詞 dog / name / find / men 在語料中極常見, BM25只把正解段落排到#8, 稍難的改寫就會把它擠出 context; Embedding e5從語意給#5, RRF 融合後來到#2。100 題 final_query是混合題型, 在語意題是e5強, 但對於精確詞／補書名／引言題 BM25表現較好如(Q5"Musgrave Ritual", BM25 #4)。RRF 會同時考量sparse retrieve 與 dense retrieve, 兩邊的排名都越靠前, RRF 越前面, 表示這個chunk綜合下來大家都認可。

3. (10%) Fill in the table below to show your setting in this phase.

Parameter	Configuration	Rationale
prompt	 可放大看, 詳細見下方	採用Zero-shot + Rule-based constraints, 之前試過模型會把few-shot範例誤當正卻解答輸出, 再依據問題型別分段設計prompt 比較有趣的設計是, 針對可以在原文找到答案的FACT題型, 我們歸納了一些常見的錯誤, 如模型會echo題目、給一段描述而非名字、把定義題答成那個詞本身、把「在哪篇」答成敘事句...。首先先寫明不要echo, 然後設計了7條規則 FACT_SYS如下: 1. NAME → 輸出專有名詞(如: Moriarty/Lestrade/221B...) 2. ABOUT/TO WHOM → 引句中的人名(抓句尾逗號後的人名"... , Watson"→ Watson) 3. FILL-IN-BLANK → 只給缺字、不准echo 4. DEFINITION(what was X) → 給X旁的定義(baritsu→a Japanese system of wrestling), -而非return X 本身 5. IN WHICH story → 只給故事名, 不准回填題目裡的地點/人物/事件、不給敘事句 6. HOW OLD/HOW MANY/WHAT YEAR → return 數字 7. 真的不確定不要亂答 → I don't know。
k_1	0.5	succeed task 2 results
b	0.75	succeed task 2 results
chunk size	256	optimized by grid search
chunk overlap	128(50%)	optimized by grid search

prompt design in task 4:

```
META_SYS = ""
```

Role: You are a world-class expert on the Sherlock Holmes canon and its publication history and famous adaptations (e.g. the BBC series 'Sherlock').

Task: This is a general-knowledge / bibliographic / real-world question. Use the retrieved notes as reference; the answer may appear in them.

Format: Output ONLY the exact short answer on ONE line (a name, a title, a number, or a single word). No sentences, no explanation.

- FILL-IN-THE-BLANK ('complete the title/dialogue', or a ____): output ONLY the missing word(s) and never repeat a word that already appears in the question (e.g. 'The ____ of the Baskervilles' -> 'Hound').

If genuinely unsure, reply exactly: I don't know

```
"".strip()
```

```
FACT_SYS = ""
```

Role: You are a world-class expert on the Sherlock Holmes canon.

Task: Answer using the retrieved excerpts. Each excerpt is prefixed with [Story: ...]; trust only excerpts that clearly concern the SAME story / person / thing the question asks about; ignore the rest.

Format: output ONLY the canonical answer on ONE line (a name, a title, a place, a number, or one word). Never copy a whole sentence, and NEVER answer with a word that is merely echoed from the question (e.g. the quoted title word, or a place / character already named in the question).

Rules:

1. NAME questions ("what is the name of...", arch-enemy/nemesis, inspector, client, address): output the proper noun (e.g. "Moriarty", "Lestrade", "221B Baker Street"). If an excerpt only describes the person/thing, give the actual name from your own canon knowledge, not the description.

2. ABOUT / TO WHOM questions: the answer is the person named or addressed in the quote – usually a name right after a comma at the end (e.g. "..., Watson" -> Watson) – NOT the topic word of the sentence.

3. FILL-IN-THE-BLANK (a title or quote containing ____): output ONLY the missing word(s); never repeat a word that already appears in the question (e.g. "The ____ of the Baskervilles" -> "Hound").

4. DEFINITION ("what was X?"): give the definition sitting NEXT TO X, not X itself ("baritsu" -> "a Japanese system of wrestling"; "the speckled band" -> "a snake (a swamp adder)").

5. IN WHICH story/novel/work something happens or appears: output a canon story TITLE only (e.g. "The Final Problem") – never a place, character, or event that is named in the question, and never a narrative sentence.

6. HOW OLD / HOW MANY / WHAT YEAR / HOW LONG: output only the number, age, or year (e.g. "seven years older", "1881").

7. If the excerpts do not contain it, answer from your own canon knowledge. If truly unsure, reply exactly: I don't know

```
"".strip()
```

```
MC_SYS = ""
```

Role: You are a world-class expert on the entire Sherlock Holmes canon by Arthur Conan Doyle.

Task: This is a multiple-choice trivia question, but the answer choices are NOT shown to you. Using your own canon knowledge (and the reference notes), decide the single correct answer.

Format: Output ONLY the exact answer on ONE line – a story title, a name, or a short phrase – with no explanation. For a "which ... is NOT ..." question, name the one item that does not fit. If truly unsure, reply exactly: I don't know

```
"".strip()
```

```
USER_PROMPT = ""
```

```
### Question
```

```
{query}
```

```
### Retrieved Sherlock Holmes passages
```

```
{relevant_chunks}
```

```
"".strip()
```

4. (10%) Describe any additional techniques(beside example code) you applied to further improve performance.

Preprocessing

擷取書名 (Story Title Parsing): 書目錄共7176個字元, 利用 Regex(re)掃描原著目錄, 自動抓出所有的故事名稱, 並計算每個故事在文本中的起訖位置 (start, end)。這使得後續每一個 Chunk 都有一個story_tag。

Query Analyzing & Routing

名言擷取與定位 (Quote Extraction & Anchoring):

遇到長字串透過 `_NORM` index物理定位該段落。

故事偵測 (Named Entity Recognition for Titles):

若題目直接提到某個故事名稱(例如 "In The Hound of the Baskervilles..."), 以此來限縮檢索範圍 (named_story)。

Prompt 設計規則與 LLM 混合路由 (Rule-based & LLM-based Routing):

目標是將題目將拆分為META、QUOTE、FACT、MC四種題型,

先使用關鍵字(如 published, year, tv series)做第一層快速分類。

如果關鍵字判斷不出來, 動用 LLM 進行 Zero-shot 分類(呼叫 Gemma 回答 META 或 FACT), 讓檢索與 Prompt 策略可以對症下藥。

META:

定義: 這類問題不是問小說裡的情節, 而是問關於《福爾摩斯》在**「現實世界」**中的出版歷史、書目資料、作者生平、或是影視改編

判斷條件: query包含關鍵字如 `bbc, magazine, published, best-seller, how many stories, tv series, complete the title` 等。

系統處理方式:

專用 Prompt (META_SYS): 要求 LLM 扮演「出版史與改編專家」, 給出非常簡短的現實事實(人名、年份或書名)

外部知識庫注入: EXT_CHUNKS+CANON_LIST

FACT:

定義: 這類問題是針對小說內部的情節、角色設定、地點或發生的事件進行提問。

判斷條件: 當問題不屬於 META 時, 或是明確問到「In which story... (在哪一個故事中...)」。若關鍵字判斷不出來, 系統會呼叫 LLM 進行 Zero-shot 判斷 (Is this asking about REAL-WORLD facts... or IN-NOVEL story details?)。

專用 Prompt (FACT_SYS): 要求 LLM 扮演「福爾摩斯正典專家」。嚴格規定 LLM 只能根據檢索到的段落回答, 並且制定了 7 條規則 (例如: 問是誰, 就只能回答人名, 不能抄完整句等等)

ABSTAIN_TAU: 如果系統發現檢索到的文本段落, 與問題的關聯性(餘弦相似度)太低, 代表原著裡可能沒寫, 程式會攔截 LLM 直接強制輸出 "I don't know" 以避免亂回答被倒扣分。

QUOTE:

定義: 這嚴格來說屬於 FACT 的一個子分支, 但它的檢索方式完全不同。這類問題會給出一長串引號包起來的句子, 要求系統填空或是指出這句話出自哪裡。

判斷條件: 以 "包裝 且長度夠長的字串。

要找一字不差的引文, 系統會先把整本小說的標點符號跟空白去掉(變成純字母字符串_NORM), 然後直接用 find() 去做精確比對, 找出這句話在原文本的絕對位置。

強制置頂 (Pinned): 一旦找到引言出處的 Chunk, 系統會把這個段落強制排在檢索結果的第 1 名, 確保 LLM 絕對能看到原文來填空。

MC(Multiple Choice):

定義: "which/which ... is NOT ...", 但選項並沒有提供給系統看 (Hidden Options), 系統必須完全依靠自己對福爾摩斯全集的理解來「推導」出答案。

條件: 包含 which of these (以下哪個), which of the following, common between (...之間的共同點), 並且不是問特定故事。

給予大補帖 (Context 擴充): 系統不僅會進行常規檢索, 還會強制把 CANON_LIST (所有故事分類) 加上外部現實知識, 全部丟進去給 LLM。

專用 Prompt (MC_SYS): 明確告訴 LLM: 「這是一道選擇題, 但選項沒有顯示給你。請利用你的知識與筆記, 決定唯一正確的答案。如果問的是 '哪個不是', 請指出唯一不合群的項目。」

Hybrid indexing

Sparse Retrieval - BM25Okapi:

把中文jieba斷詞換成英文Tokenizer, 使用 NLTK 進行停用詞 (Stopwords) 過濾與詞幹提取 (PorterStemmer), 建立基於關鍵字頻率的 BM25 索引。

Dense Retrieval - Embedding:

將所有 Chunk 轉為向量並進行 L2 正規化 (Normalization), 以便後續直接使用矩陣內積 (Dot Product) 快速計算餘弦相似度 (Cosine Similarity)。

External Knowledge Injection (提升最大)

According to the QA spaces, it's allowed to add External knowledge chunks

CANON_LIST:

LLM常分不清出版書籍先後與誰是誰的續集等等, 透過預先照順序建好CANON_LIST, LLM得以知道書籍或章節之間的關係

EXT_CHUNKS:

lots of answers cannot be found in SherlockHolmesComplete.txt

手動將Wikipedia 風格的事實句子 (有關福爾摩斯的作者生平、出版年份、BBC 影集改編...)直接加進檢索池中, 解決了純文本檢索無法回答 Meta Question 的痛點。

Peer Evaluation

For the peer evaluation table, each group may rate member contributions on a 1-10 scale. You may decide the score based on your own group collaboration.

Member	Student ID	Name	Score(1-10)	Contribution
01	113550020	賴宇弘	10	Task 1 Simple Baseline: LLM Only
02	112705068	何昀珍	10	Task 2 Medium: BM25 + LLM
03	112701018	姜明志	10	Task 3 Strong: embedding + LLM
04	111701030	程品諭	10	Task 4 Hybrid: LLM+BM25+embedding

Report Submission Format Requirements

File Format:

- Submit your result, code, and report to E3 **separately**.
- Please note that **each group** should **upload only one set** of files in total.
- **Report:** Submit in **.pdf** format.
 - **Filename:** `CLASSNUMBER_GROUPNUMBER_HW4_report.pdf`
 - example: `515512_Group01_HW4_report.pdf`
 - Please refer to E3 to check if you're from class **515512** or **515513**.
 - **Please make sure to include all members' student IDs and names at the very beginning of the report.**

Grading Policy:

- Wrong submission format leads to 0 points.
- Late submission leads to 0 points.
- Missing student ids and names in the report leads to 0 points